Story Point Estimation with BERT Word Embeddings - Final Report

PS-3: Evaluate BERT-based word embeddings for story point estimation when training on source projects and testing on a different target project under the assigned random-split protocol.

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GitHub: https://github.com/ettasrinith/story-point-estimation

#1. Problem

Agile teams label each issue with a story point - a rough measure of the work it needs. We study whether a model that reads the issue text can predict this number, and specifically whether such a model transfers across projects: train on one project's issues, test on a different project it never saw. This is repeated over the 39 training/testing pairs assigned in the project list, using BERT word embeddings as the text representation, a three-layer CNN as the predictor, random 80:20 splitting, and MAE / MdAE as the metrics.

#2. Dataset

TAWOS (Tawosi et al., MSR 2022; DOI 10.5522/04/21308124) - 65427 Jira issues with story points across 39 open-source projects (extracted from the official MySQL dump by 01_extract_tawos.ipynb; per-project counts in results/manifest.csv).

Cleaning: story points cast to numeric, issues without a story point dropped (458,232 -> 65,427), HTML markup stripped from title + description, duplicates removed. For all experiments story points are kept in (0, 40] - this removes zero-effort tickets and broken values (e.g. one project contains a story point of 26,710,114) - leaving 62,193 issues. Project sizes range from 209 (Apache MXNet) to 19,358 (Lsstcorp Data management) issues.

#3. Method

Representation. bert-base-uncased, frozen, inference only. Each issue's text (title + description) is tokenized (max length 128, ~13% of the longest descriptions truncated) and encoded once; the token-level last-hidden-state vectors [128 × 768] are stored per project (float16, 12.2 GB total; 03_bert_embeddings.ipynb).

Model. Three-layer 1D CNN (768->256->128->64 channels, kernel 3, ReLU, dropout 0.2 on the first layer) over the token embeddings, masked global max pooling over real tokens, then a small MLP head (64->32->1) predicting the story point. L1 loss (matches MAE), Adam lr 1e-3, batch 64, early stopping on validation MAE (patience 4, max 25 epochs). fp16 autocast on a T4 GPU.

Protocol (per pair). Each project's issues are split randomly 80:20 with seed 42 - the same partitions used by the baseline, so results are directly comparable. (1) 5-fold cross-validation on the source project's 80% pool: train on 4 folds, validate on 1; reports mean +/- std validation MAE.

(2) The number of epochs is fixed to the folds' average, and a final model is trained on the full pool. (3) Single evaluation on the target project's held-out 20%. The target project's data never touches training or model selection.

Baseline. Predict the source project's median story point for every target issue (02_baseline_median.ipynb).

Two project names in the assignment table have no exact match in the dataset ("Evergreen", "C++ driver (Appcelerator)"); with the instructor's approval they are mapped to the two remaining TAWOS projects, "The MongoDB Engineering" and "Aptana Studio".

#4. Results

Main model: pooled-source BERT+CNN (06). Following GPT2SP (Fu & Tantithamthavorn, 2023), each target model trains on the combined train pools of ALL other projects instead of a single source (target data fully excluded), the network learns log1p(story point) to handle the heavy skew, and 3 seeds are averaged. A global 5-fold CV on the pooled training data selects the number of epochs.

Model	Test MAE (mean +/- std over pairs)	Test MdAE	Beats baseline
Median baseline	2.496 +/- 1.450	1.872	-
Per-source BERT+CNN (05)	2.326 +/- 1.350	1.710	27/39
Pooled-source BERT+CNN (06)	2.024 +/- 1.212	1.351	29/39

The pooled model reduces test MAE by 18.9% over the baseline and 13.0% over the per-source model. Biggest gains come from pairs where the single source was small (e.g. Appcelerator Studio -> Nexus: 4.241 -> 1.019; Blockchain Explorer -> Moodle: 1.619 -> 1.464), because the pooled model no longer depends on the vocabulary of one small project.

Per-pair pooled results: results/results_pooled_pairs.csv (plot: results/results_pooled_per_pair.png).

##4.1 Per-source model (ablation, all 39 assigned pairs)

Metric	BERT + CNN	Median baseline
Test MAE (mean +/- std over pairs)	2.326 +/- 1.350	2.496 +/- 1.450
Test MdAE (mean over pairs)	1.710	1.872
Mean 5-fold validation MAE	1.923 +/- 0.213	-
Pairs where CNN beats baseline	27/39	-

Per-pair results (validation from 5-fold CV on the source; test = held-out target):

Source	Target	Val MAE (5-fold)	Test MAE	Test MdAE	Baseline MAE

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XD    MESOS    1.923 +/- 0.027    1.572    1.139    1.538
Usergrid    MXNet    0.887 +/- 0.103    1.582    0.942    2.425
Titanium Mobile Platform    Usergrid    2.363 +/- 0.308    2.346    1.688    0.729
The Titanium SDK    Alloy Framework    2.948 +/- 0.212    1.757    1.419    2.180
Software Cloud    Appcelerator Daemon    2.339 +/- 0.203    7.176    4.598    3.375
Sawtooth    Appcelerator Studio    0.899 +/- 0.084    3.231    2.866    3.646
Nexus    Command-Line Interface    0.973 +/- 0.048    1.874    1.191    2.789
MXNet    The Titanium SDK    3.281 +/- 0.908    3.149    1.558    3.470
Mule APIkit    Titanium Mobile Platform    1.782 +/- 0.304    4.287    3.065    3.168
Mule    C++ driver (Appcelerator)    2.798 +/- 0.136    3.765    2.590    5.880
Moodle    Atlassian Software Server    7.074 +/- 0.433    4.930    3.354    2.492
MongoDB Core Server    Bamboo    1.402 +/- 0.186    0.901    0.692    1.277
MESOS    Clover    1.476 +/- 0.121    3.693    1.374    4.178
Lyrasis Dura Cloud    Confluence Cloud    1.105 +/- 0.232    2.025    2.023    2.390
Lsstcorp Data management    Confluence Server    3.265 +/- 0.050    1.333    1.098    1.039
Jira Server    Crowd    1.461 +/- 0.226    1.586    1.753    2.558
Jira Cloud    FishEye    1.337 +/- 0.215    1.478    1.484    2.169
Java driver    Jira Cloud    0.085 +/- 0.050    2.990    3.131    3.243
Indy SDK    Jira Server    1.528 +/- 0.087    3.613    1.189    1.987
Indy Node    Software Cloud    1.394 +/- 0.148    1.503    1.181    2.237
FishEye    DotNetNuke Platform    1.037 +/- 0.138    0.890    0.633    1.040
Fabric    Blockchain Explorer    1.654 +/- 0.101    1.176    0.900    1.501
Evergreen    Fabric    0.756 +/- 0.031    0.970    0.821    1.752
DotNetNuke Platform    Indy Node    0.900 +/- 0.044    1.699    1.066    1.956
DataCass    Indy SDK    2.403 +/- 0.411    2.383    1.916    1.620
Crowd    Sawtooth    0.845 +/- 0.108    1.165    0.729    1.368
Confluence Server    Lsstcorp Data management    1.306 +/- 0.113    3.177    1.540    3.886
Confluence Cloud    Lyrasis Dura Cloud    1.325 +/- 0.196    1.051    0.984    1.878
Compass    MongoDB Core Server    1.374 +/- 0.127    1.243    1.133    1.490
Command-Line Interface    Compass    1.842 +/- 0.219    1.253    1.013    1.300
Clover    C++ driver (MongoDB)    3.016 +/- 0.334    2.122    1.784    0.917
C++ driver (Appcelerator)    Evergreen    3.965 +/- 0.399    3.567    3.403    6.099
C++ driver (MongoDB)    Java driver    0.482 +/- 0.102    0.549    0.103    0.000
Blockchain Explorer    Moodle    1.226 +/- 0.173    1.619    1.392    6.858
Bamboo    Mule    1.309 +/- 0.237    2.285    1.279    3.150
Atlassian Software Server    Mule APIkit    2.281 +/- 0.213    1.872    1.537    1.882
Appcelerator Studio    Nexus    2.245 +/- 0.051    4.241    4.152    3.390
Appcelerator Daemon    DataCass    4.899 +/- 0.960    2.579    2.141    2.401
Alloy Framework    XD    1.798 +/- 0.259    2.066    1.852    2.079
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Best pairs: C++ driver (MongoDB)->Java driver (0.549), FishEye->DotNetNuke Platform (0.890), MongoDB Core Server->Bamboo (0.901).
Worst pairs: Software Cloud->Appcelerator Daemon (7.176), Moodle->Atlassian Software Server (4.930), Mule APIkit->Titanium Mobile Platform (4.287).

Demo pair walkthrough (Nexus -> Command-Line Interface, 04_cnn_model.ipynb): 5-fold val MAE 0.973 +/- 0.048, test MAE 1.874 vs baseline 2.789.

#5. Discussion

- **The per-source CNN beats the median baseline on 27 of 39 pairs (test MAE 2.496 ->** 2.326), and the pooled-source model improves this further to 2.024 - BERT word embeddings clearly carry effort-relevant signal that transfers across projects, and pooling sources multiplies the usable training data per target.
- **Per-source validation MAE (~1.92) is much lower than its test MAE: the model is** accurate *within* the source project but degrades on unseen projects. The transfer gap is the central finding of this setting. Notably, the pooled model's global 5-fold validation MAE (2.024-scale) is less informative per pair - pooled validation mixes many projects' scales, so per-pair test numbers remain the primary evidence.
- **Worst pairs combine a source and target with very different story-point scales (e.g. a source with** median 1-2 and a target with median 5); text alone cannot recover the target's scale, so the model inherits a systematic offset. Best pairs share vocabulary domains and scales.
- **Validation results are reported separately from held-out test results, per §3.4; the final** configuration (epochs) was selected using development folds only.

#6. Threats to validity

- **Construct: story points are team-specific estimates, not objective effort; the (0, 40] filter** and per-project scales make cross-project MAE hard to interpret in absolute terms.
- **Internal: fixed seeds and partitions for all methods; model selection used development folds** only; the target project never influenced training. The two instructor-approved name mappings ("Evergreen" -> "The MongoDB Engineering", "C++ driver (Appcelerator)" -> "Aptana Studio") are the only interpretive steps in the pair list.
- **External: 39 pairs from one dataset (open-source Jira projects); results may not generalize to** commercial trackers or other estimation scales.
- **Statistical: one run per pair with a fixed seed; fold variance is reported, but no repeated** runs with different global seeds.

#7. Conclusion

BERT word embeddings with a small CNN improve cross-project story point estimation over a median baseline on 27/39 assigned pairs (test MAE 2.326 vs 2.496), but a large project-transfer gap remains. Pooling source projects, a log target and seed ensembling cut test MAE to 2.024; scale mismatch between projects is the dominant remaining obstacle - combining text with target-project calibration is the natural next step.

#Reproducibility

All notebooks run on Kaggle: 01 (data extraction, CPU), 02 (baseline, CPU), 03 (embeddings, T4), 04 (demo pair, T4), 05 (per-source 39 pairs, T4), 06 (pooled model, T4, ~2 h). Seeds, split procedure and configurations are fixed inside the notebooks; results CSVs are committed under results/.